

Week 4: UI Layout Development

Objective

Develop the main graphical user interface layout of the Scientific Calculator using Java Swing. This week focuses on the header, display panel, status area, and overall visual theme.

Tasks

1. Develop Header Section

- Create the calculator header using `JPanel`.
- Add the title **SCIENTIFIC CALCULATOR**.
- Add course/student information.
- Add the ON/OFF power button.
- Apply fonts, colors, spacing, and alignment.

2. Create Display Panel and Status Area

- Create the main display panel.
- Add the main calculator display.
- Add the power/status indicator.
- Align display information to create a clean calculator interface.

3. Apply Theme and Styling

- Apply a dark background theme.
- Define primary and secondary text colors.
- Apply accent colors for different UI elements.
- Configure fonts and borders.
- Apply consistent spacing and alignment.

Code Portions Used

Theme Colors

```
private final Color BG_DARK = new Color(24, 24, 27);
private final Color BG_PANEL = new Color(39, 39, 42);
private final Color TEXT_MAIN = new Color(250, 250, 250);
private final Color TEXT_MUTED = new Color(161, 161, 170);
private final Color ACCENT_BLUE = new Color(14, 165, 233);
private final Color ACCENT_RED = new Color(239, 68, 68);
private final Color ACCENT_GREEN = new Color(34, 197, 94);
private final Color ACCENT_ORANGE = new Color(249, 115, 22);
```

Header

The `buildHeader()` method creates the title, subtitle, and power button and applies their styling.

Display and Status

The `buildDisplay()` method creates the history label, main display, and status label.

Main Layout

```
setLayout(new BorderLayout(0, 0));
getContentPane().setBackground(BG_DARK);

add(buildHeader(), BorderLayout.NORTH);
add(buildDisplay(), BorderLayout.CENTER);
add(buildButtons(), BorderLayout.SOUTH);
```

Week 4 Deliverables

- Header section completed
- Display panel completed
- Status area completed
- Dark theme applied
- Fonts and colors configured